Problem Understanding

Team name

BLACK GOLD

Objective

Predict permeability from basic well logs to create a continuous permeability profile for reservoir evaluation.

Challenge

- Permeability is highly variable, non-linear, and difficult to measure.

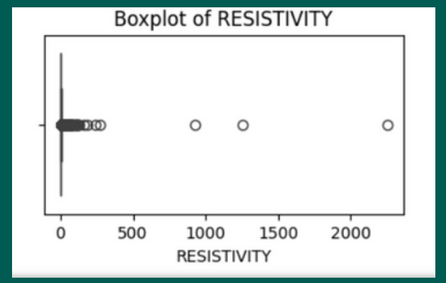
Input Logs

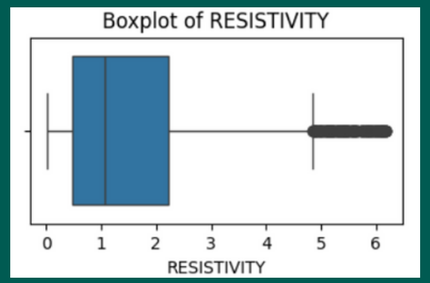
- Gamma Ray (GR) – lithology indicator
- Resistivity (RT) – fluid/lithology response
- Porosity logs – pore space information

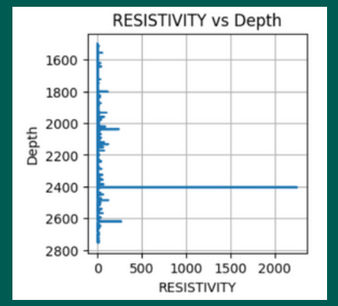
Output

Permeability (k) – How easily fluids flow through the rock.

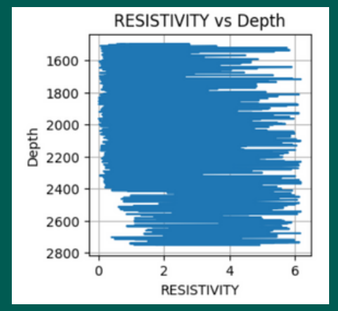
“Workflow and Methodology

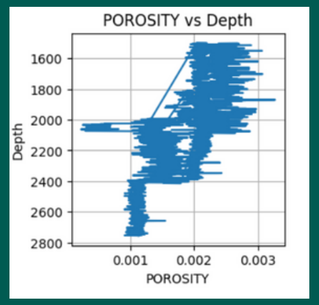


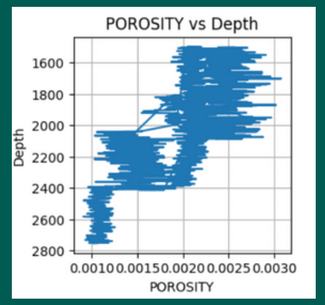




After removing outliers







Include:

1.**Data loading (from 4 well logs)** - las files to dataframes

2.**QC & Filtering (missing values, outlier removal)** - used IQR method.

3.**Exploratory Data Analysis (EDA)-** made histogram and box plot for every Gamma, Porosity, Perm and Resistivity.

4.Also made depth vs gamma/porosity/perm/resistivity plots

4.**Feature Engineering**

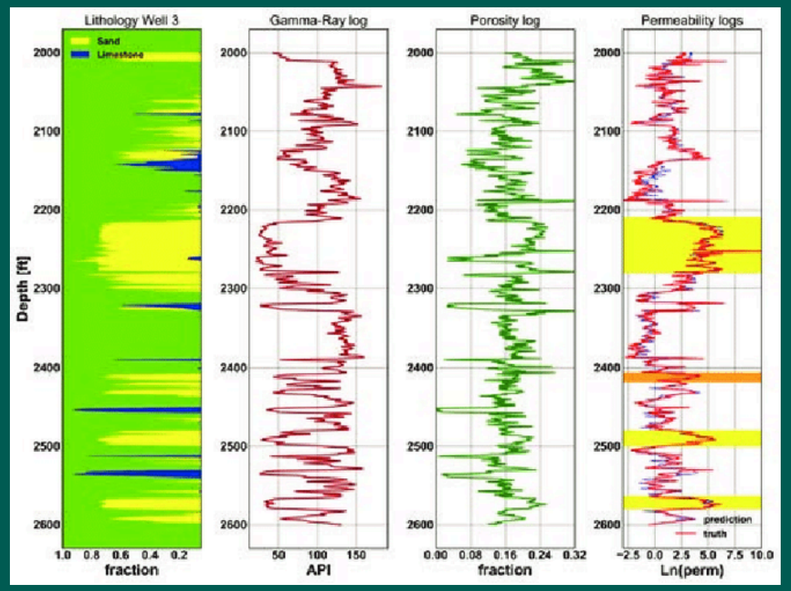
5.**Model training & validation**

6.**Test on blind well (**“test.las”**)**

7.**Final evaluation**

Feature Engineering

- **Outlier removal** using IQR
- **Normalization / standardization** (why? → resistivity & porosity have different scales) using **StandardScaler()** for numerical columns and **OneHotEncoding()** for categorical columns
- **Derived features**:
  - Log(resistivity)
  - Shale volume from GR
- **Handling missing values** - removed
- **Train/Validation split**




```python
df_train['log_RES'] = np.log(df_train['RESISTIVITY'] + 1)
GR_min = df_train['GAMMA'].quantile(0.05)
GR_max = df_train['GAMMA'].quantile(0.95)

df_train['VSH'] = (df_train['GAMMA'] - GR_min) / (GR_max - GR_min)
df_train['VSH'] = df_train['VSH'].clip(0, 1)
df_train.head()
```

METHOD AND MODEL SELECTION

Random Forest

model = RandomForestRegressor(n_estimators=200, random_state=42)

Using Pipeline and ColumnTransformer

LOGO METHOD

Leave-One-Group-Out Cross-Validation

Fold	Training Wells	Testing Well
1	A, B, C	D
2	A, C, D	B
3	B, C, D	A
4	A, B, D	C

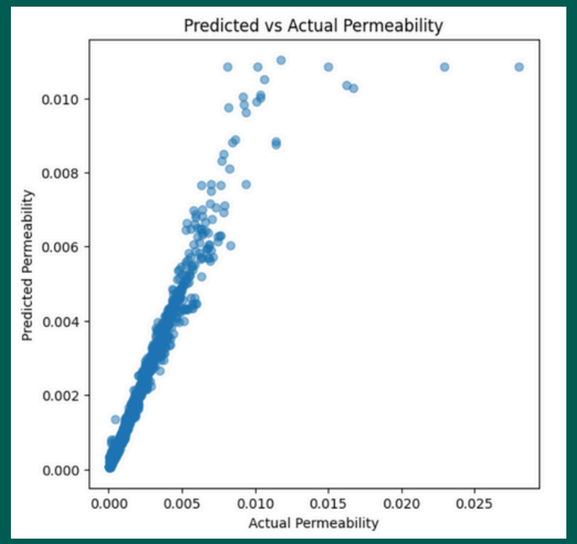
Benefits of LOGO

- Eliminates well-to-well data leakage
- Provides realistic generalization performance
- Suitable for small datasets with strong grouping pattern
- Useful accuracy metrics for deployment in new wells

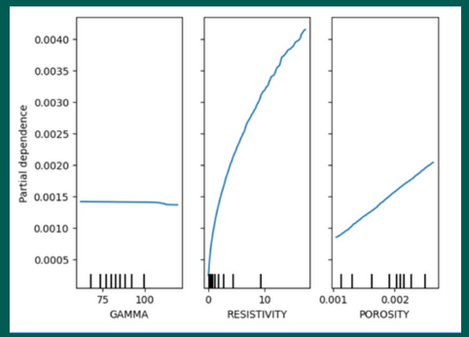
```
for train_idx, test_idx in logo.split(X_train_f, y_train_f, groups):
    well_name = df_train.iloc[test_idx]['WELL'].unique()[0]


    X_train, X_test = X_train_f.iloc[train_idx], X_train_f.iloc[test_idx]
    y_train, y_test = y_train_f.iloc[train_idx], y_train_f.iloc[test_idx]
```

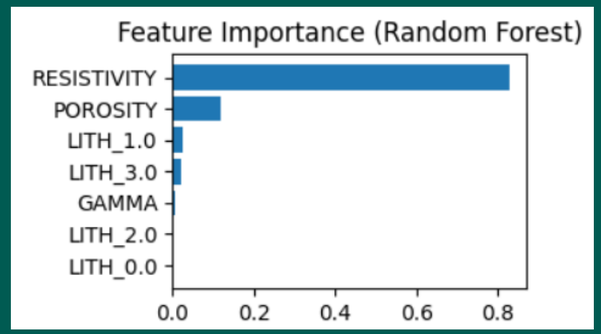
Results & Kaggle Score

Predicted vs Actual Permeability Plot



Sentivity analysis





Model Performance Metrics

Mean ± Standard deviation

```
RMSE: 0.0002 ± 0.0001
MAE: 0.0001 ± 0.0000
R2: 0.9561 ± 0.0417
```

	id	Permeability
1	1902.2873	0.000830142856
2	1902.7873	0.0008288557419999985
3	1903.2873	0.001049198848
4	1903.7873	0.0014356169004999997
5	1904.2873	0.0016902166139999982
6	1904.7873	0.0018296816809999982
7	1905.2873	0.001968679113000001
8	1905.7873	0.0020386327674999966
9	1906.2873	0.0019465992135000026
10	1906.7873	0.0017367784890000022
11	1907.2873	0.0016303072675000025
12	1907.7873	0.0015346507315000008
13	1908.2873	0.001322182173
14	1908.7873	0.001009956855500001
15	1909.2873	0.0007265669569999986
16	1909.7873	.00046252193799999937
17	1910.2873	.00027901098599999964
18	1910.7873	.00019242638900000014
19	1911.2873	0.0001844038064999999
20	1911.7873	0.0002322195065000002
21	1912.2873	.00028885251599999976
22	1912.7873	0.0003598863429999997

Key insights + What you would improve

Key Insights

- Resistivity (RT) was the most influential feature in predicting permeability.
- The model generalized well to the blind well, showing consistent predicted vs. actual trends.
- Feature engineering significantly improved performance, leading to lower RMSE and more stable predictions.

Improvements

Add More Feature Engineering (Huge Accuracy Boost)

- Compute gradients (dGR/dDepth, dRes/dDepth)
- Resistivity ratios (deep/shallow)

Try More Advanced Models

- XGBoost, CatBoost
- Neural Networks (if you have enough data)
- Then compare performance using LOWO.

Use GridSearchCV / RandomizedSearchCV to tune

Thank You